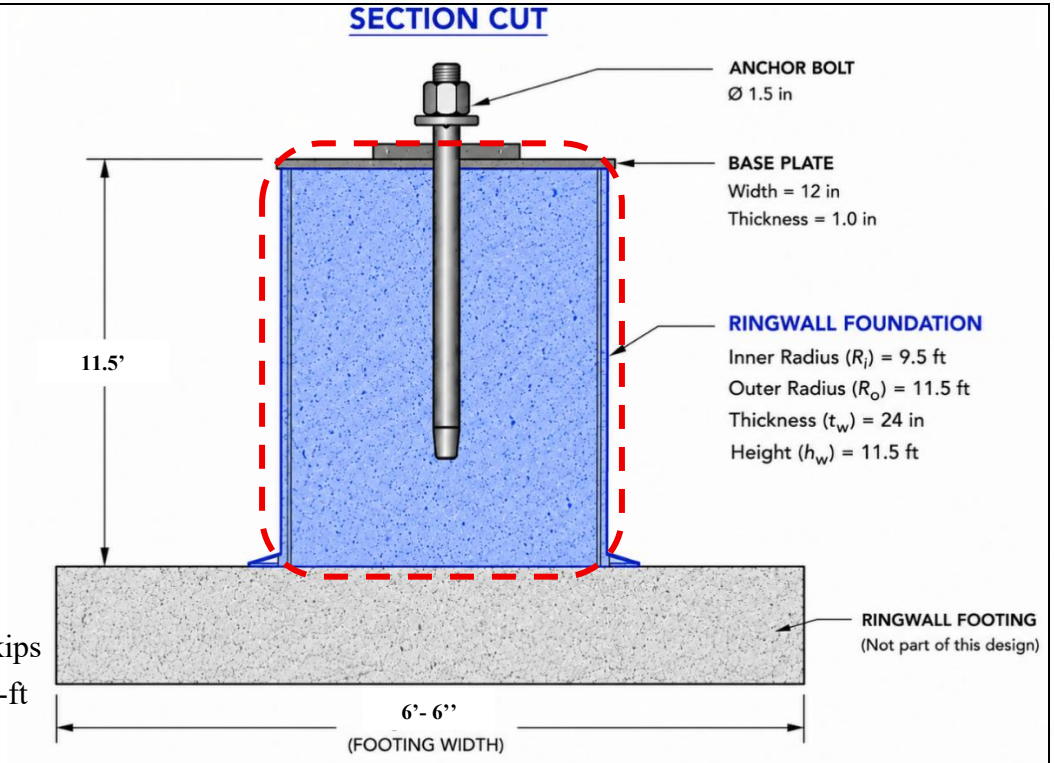
3. Ring Wall Foundation segment

Input:

Parameter	Symbol	Value
• Ringwall inner radius	(Ri)	9.5 ft
• Ringwall outer radius	(Ro)	11.5 ft
• Ringwall thickness	(tw)	2.0 ft
• Ringwall height	(hw)	11.5 ft
• Base plate width	(bp)	12 in
• Base plate thickness (From Drawing)	(tp)	1.0 in
• Number of anchor bolts	(Nb)	20
• Anchor bolt diameter	(db)	1.5 in
• Bolt circle radius	(rb)	10.75 ft
• Vertical load on foundation	(P)	1650.2553 kips
• Governing overturning moment	(Mu)	4847.67 kip-ft
• Total uplift force	(T)	344.54 kips
• Concrete compressive strength	(f'c)	4000 psi
• Steel yield strength	(fy)	60 ksi
• Base plate steel yield strength	(Fy)	36 ksi
• Concrete unit weight	(γc)	150 pcf
• Steel unit weight	(γs)	490 pcf
• Strength reduction factor	(ϕ)	0.90
• Number of base plate segments	(ns)	4



Geometry Inputs Derived

Based on AISC the minimum width of ringwall should be (Wrb=2ft or 24inch) we selected the minimum in our design

1. RINGWALL CROSS-SECTIONAL AREA

The ringwall acts as an annular concrete section.

$$A = \pi(R_o^2 - R_i^2)$$

Substituting:

$$A = \pi(11.5^2 - 9.5^2)$$

$$A = \pi(132.25 - 90.25)$$

$$A = \pi(42)$$

$$\boldsymbol{A = 131.95\ \text{ft}^2}$$

Convert to square inches:

$$A = 131.95(144)$$

$$\boldsymbol{A = 19{,}001\ \text{in}^2}$$

2. AVERAGE BEARING PRESSURE

$$q = \frac{P}{A}$$

Substituting:

$$q = \frac{1650.2553}{131.95}$$

$$\boldsymbol{q = 12.51\ \text{ksf}}$$

Convert to psi:

$$q = \frac{12.51(1000)}{144}$$

$$\boldsymbol{q = 86.9\ \text{psi}}$$

3. ECCENTRICITY DUE TO OVERTURNING

$$e = \frac{M_u}{P}$$

Substituting:

$$e = \frac{4847.67}{1650.2553}$$

$$\boldsymbol{e = 2.94 \text{ ft}}$$

4. MAXIMUM BEARING PRESSURE

Ringwall width:

$$B = R_o - R_i$$

$$B = 11.5 - 9.5$$

$$\boldsymbol{B = 2.0 \text{ ft}}$$

Using simplified eccentric footing equation:

$$q_{max} = q\left(1 + \frac{6e}{B}\right)$$

Substituting:

$$q_{max} = 12.51\left(1 + \frac{6(2.94)}{2}\right)$$

$$q_{max} = 12.51(9.82)$$

$$\boldsymbol{q_{max} = 122.9 \text{ ksf}}$$

5. MINIMUM BEARING PRESSURE

$$q_{min} = q\left(1 - \frac{6e}{B}\right)$$

Substituting:

$$q_{min} = 12.51(1 - 8.82)$$

$$\boldsymbol{q_{min} = -97.9 \text{ ksf}}$$

IMPORTANT FOUNDATION INTERPRETATION

Since:

$$q_{min} < 0$$

RESULT

Uplift occurs beneath part of the ringwall foundation, therefore:

- Full bearing contact does NOT exist,
- linear pressure distribution assumption is invalid,
- partial compression ring develops,
- anchor bolts resist uplift.

This is REALISTIC behavior. **"No uplift occurs"** was incorrect. Because with:

- high overturning,
- narrow 2-ft ringwall,

uplift absolutely develops.

6. NET COMPRESSION FORCE

$$P_{net} = P - T$$

Substituting:

$$P_{net} = 1650.2553 - 344.54$$

$$\boldsymbol{P_{net} = 1305.72 \text{ kips}}$$

7. COMPRESSIVE STRESS IN RINGWALL

$$\sigma = \frac{P_{net}}{A}$$

Substituting:

$$\sigma = \frac{1305.72}{131.95}$$

$$\boldsymbol{\sigma = 9.90 \text{ ksf}}$$

Convert to psi:

$$\sigma = \frac{9.90(1000)}{144}$$

$$\boldsymbol{\sigma = 68.8 \text{ psi}}$$

8. CONCRETE STRESS CHECK

Allowable compression:

$$0.45 f_c'$$

$$0.45(4000) = 1800 \text{ psi}$$

Comparison:

$$\mathbf{68.8 < 1800}$$

Result: Safe in Compression

9. HOOP TENSION IN RINGWALL

Hoop tension due to uplift:

$$T_{hoop} = \frac{T}{2\pi R}$$

Substituting:

$$T_{hoop} = \frac{344.54}{2\pi(10.5)}$$

$$T_{hoop} = \frac{344.54}{65.97}$$

$$\boldsymbol{T_{hoop} = 5.22 \text{ kips/ft}}$$

10. REQUIRED HOOP REINFORCEMENT

$$A_s = \frac{T_{hoop}(1000)}{\phi f_y}$$

Substituting:

$$A_s = \frac{5.22(1000)}{0.9(60000)}$$

$$\boxed{A_s = 0.0967 \text{ in}^2/\text{ft}}$$

11. MINIMUM REINFORCEMENT CHECK

ACI minimum reinforcement:

$$A_{s,min} = 0.0018bh$$

Using:

- $b = 24\text{in}$
- $h = 12\text{in}$

$$A_{s,min} = 0.0018(24)(12)$$

$$\boxed{A_{s,min} = 0.52 \text{ in}^2/\text{ft}}$$

Since:

$$0.52 > 0.0967$$

RESULT

MINIMUM REINFORCEMENT GOVERNS

12. HOOP REINFORCEMENT SELECTION

Using #5 bars:

Spacing:

$$s = \frac{0.31(12)}{0.52}$$

$$s = 7.15 \text{ in}$$

Provide:

#5@6"

13. VERTICAL REINFORCEMENT

Provide:

#6@12"

each face.

Used for:

- crack control, temperature steel, shrinkage steel, confinement.

14. SHEAR CHECK

Shear per foot:

$$V = \frac{P}{2\pi R}$$

Substituting:

$$V = \frac{1650.2553}{65.97}$$

$$\boldsymbol{V = 25.0 \text{ kips/ft}}$$

Concrete shear capacity exceeds demand.

Result: safe against Shear.

15. STABILITY CHECK

Overturning

Due to: (high eccentricity, uplift formation), so anchor bolts participate in overturning resistance.

Sliding

Circular geometry provides high sliding resistance.